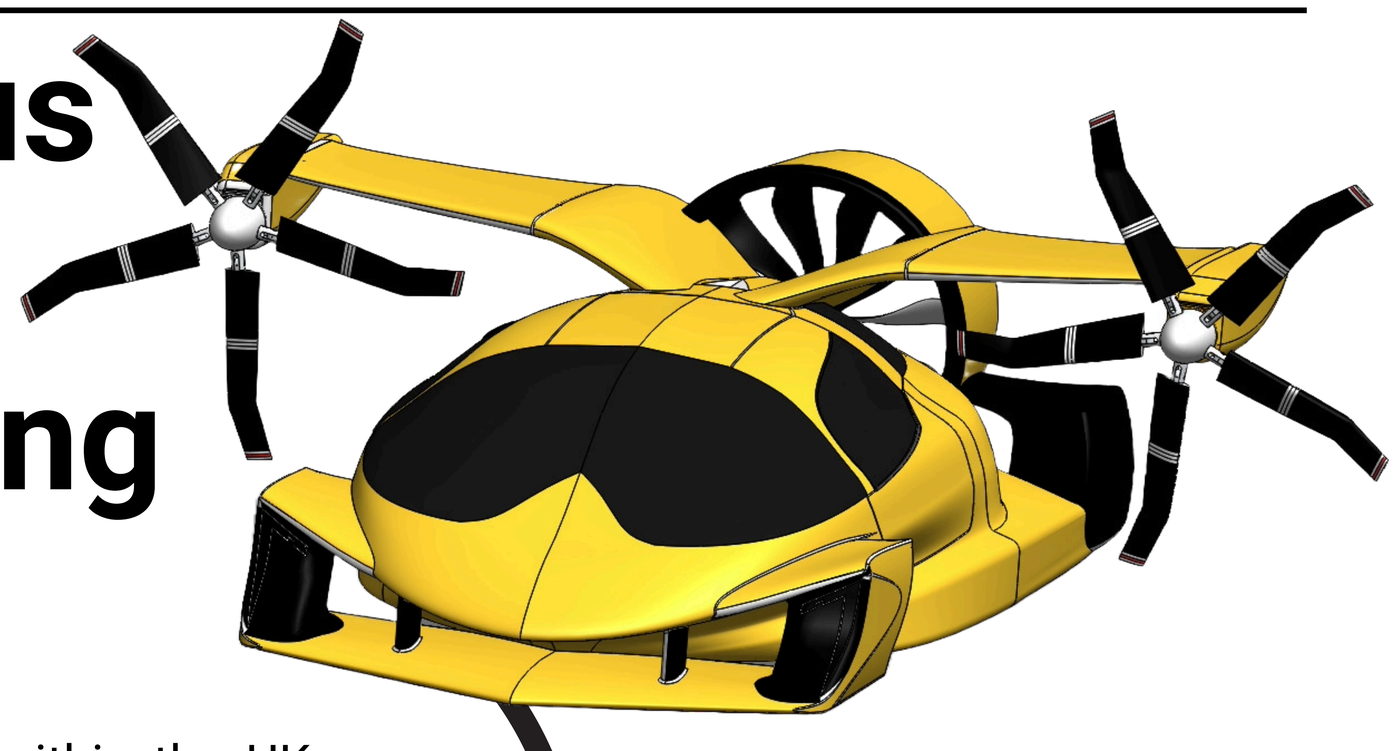
By Marisa Allen[xxtl87], Luke Harvey[nttk67], Rory Hlustik[fwvp47], Ana Nila[hpth75], Ciaran Reed [vcqt81], Luke Treadwell[sfgr47] and Shin Yoshizawa[mxyk17].

Are autonomous flying vehicles capable of saving lives?



Over **17,000** landslides have been recorded within the UK. These, with other freak hazardous events like flooding have devastating consequences, both for transport infrastructure and lives. Providing tarted bases for autonomous flying disaster relief vehicles in hotspots, E.G Cumbria, could **save lives**!

1

Motivation

- a key separating factor for the 2004 Boscastle and 1952 Lynmouth floods was the **availability of helicopters.**
- With climate change, freak nature events occur with ever increasing frequency. This **strains** the current emergency response capacity

2

Methodology

- To develop autonomous navigation system based on: Global Navigation Satellite System (GNSS), Camera and Inertial Measurmen Unit(IMU).
- Communication-based digital twin system
- Develop a vehicle with carbon fiber and titanium.
- The aim is a vehicle powered by batteries and solar panels.

3

Tech gap

Helicopter technology is currently **underutilized**. With the latest developments, our solution would combine existing aviation technology alongside cutting edge autonomous vehicle innovation. Providing unmatched affordable and reliable aviation solutions for the general public.

4

Further steps

- Mass integration with Disaster Relief policy for reduced-access locations
- Education programs to guide and inform the public on their use and justification

5

Conclusion

Flying autonomous vehicles have potential to **transform** the current transport and disaster relief infrastructure. Boosting accessibility in traditionally restrictive locations and potentially saving lives.

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